FUTURE INTERNSHIP
2026

EMAIL ANALYSIS REPORT

INTRODUCTION

This report presents an analysis of a test email generated using GoPhish and delivered via Gmail SMTP servers. The objective is to understand email structure, header information, authentication mechanisms (SPF, DKIM, DMARC), and identify any security gaps. This type of analysis is essential in cybersecurity, especially for detecting phishing attacks and verifying email authenticity.

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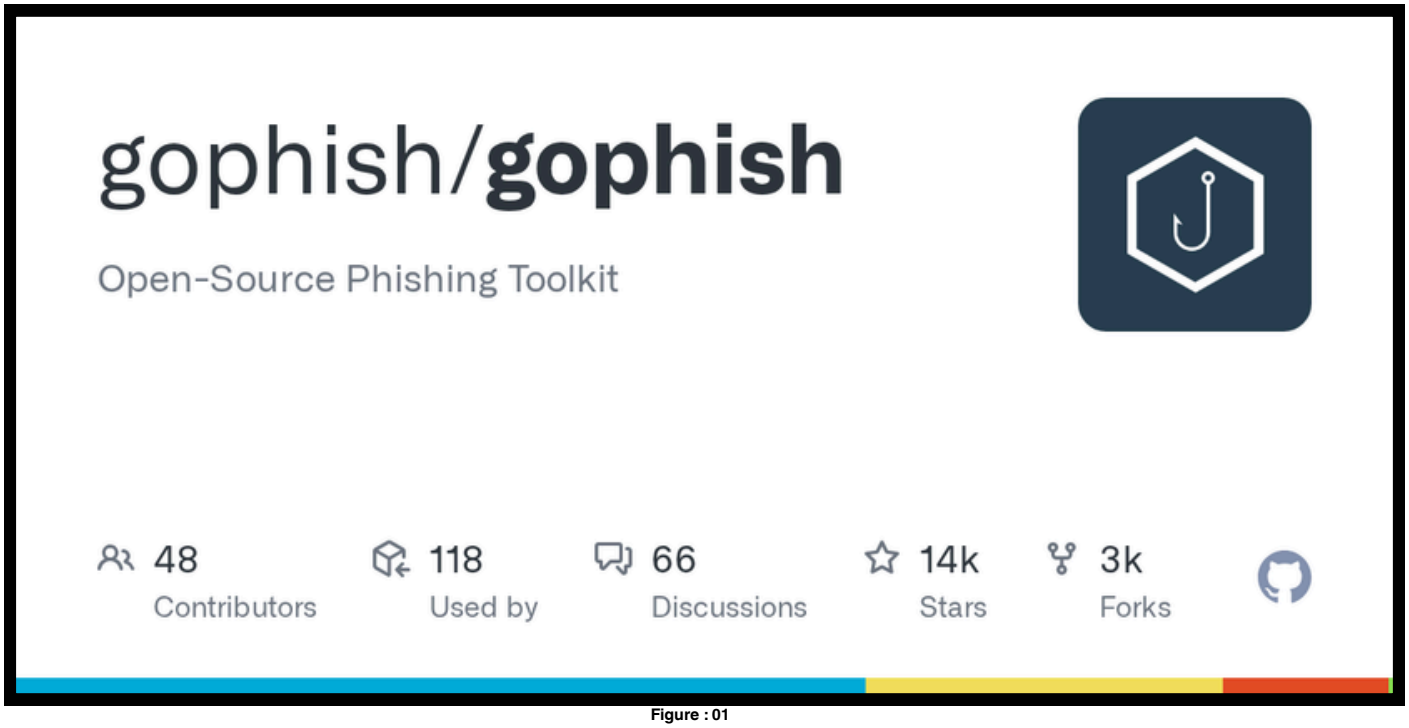

Figure : 01

ABOUT THE TOOL

GoPhish is an open-source phishing simulation platform widely used by security professionals to test and train users against phishing attacks.

Key Features:

- Campaign creation and email template design
- SMTP integration (e.g., Gmail SMTP)
- Tracking email opens, clicks, and submissions
- Generating reports for awareness training

In this case, GoPhish was configured successfully and sent a test email confirming proper setup.

EMAIL ANALYSIS

Email Overview

- Subject: Default Email from GoPhish
- From: Fakemail@gmail.com
- To: **********@gmail.com
- Mailer: gophish
- Content: Plain text confirmation email

This is a self-test email confirming that the phishing setup works correctly.

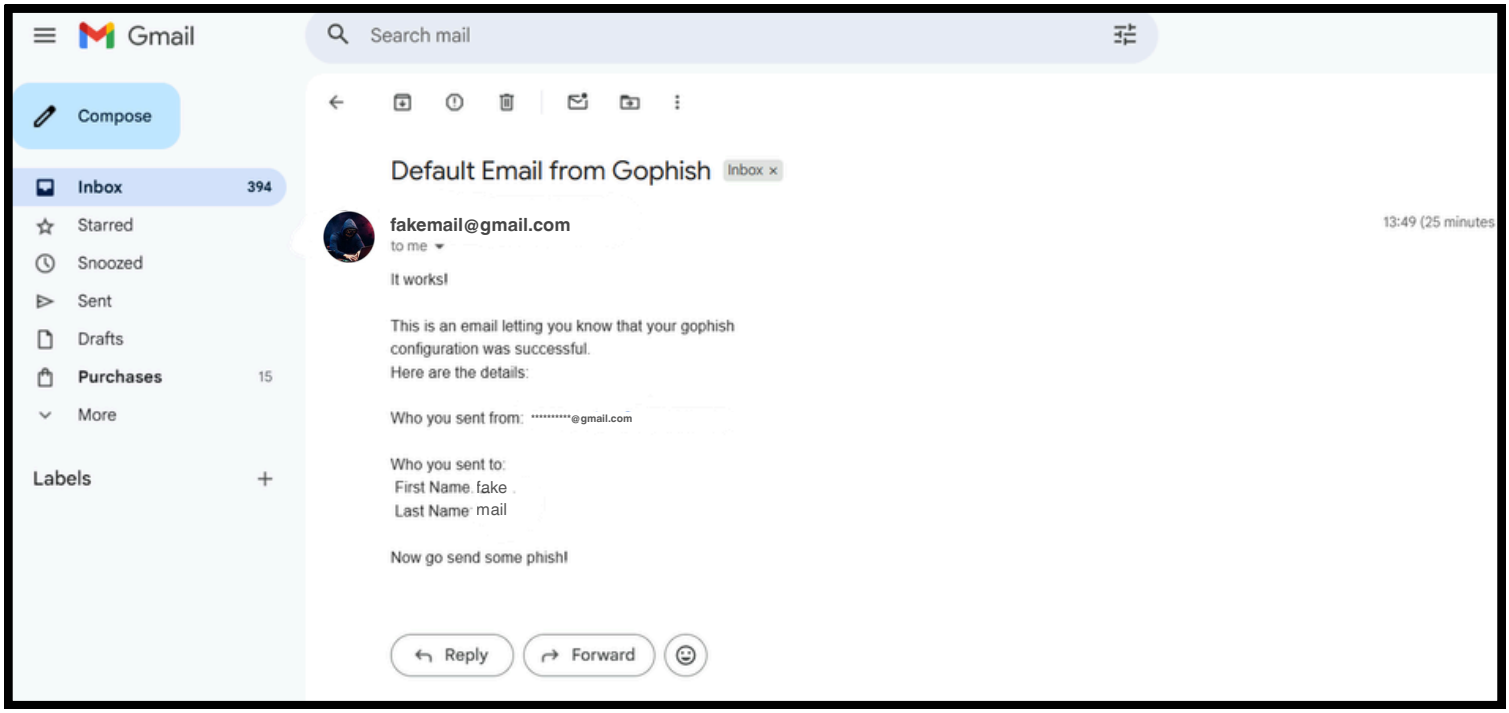

Figure : 02

HEADER ANALYSIS

Key header fields identified:

- Return-Path: *********@gmail.com
- Message-ID: Generated by Gmail mail server
- X-Mailer: gophish
- Content-Type: text/plain

The presence of X-Mailer: gophish clearly indicates the email source tool.

Headers are consistent and not spoofed.

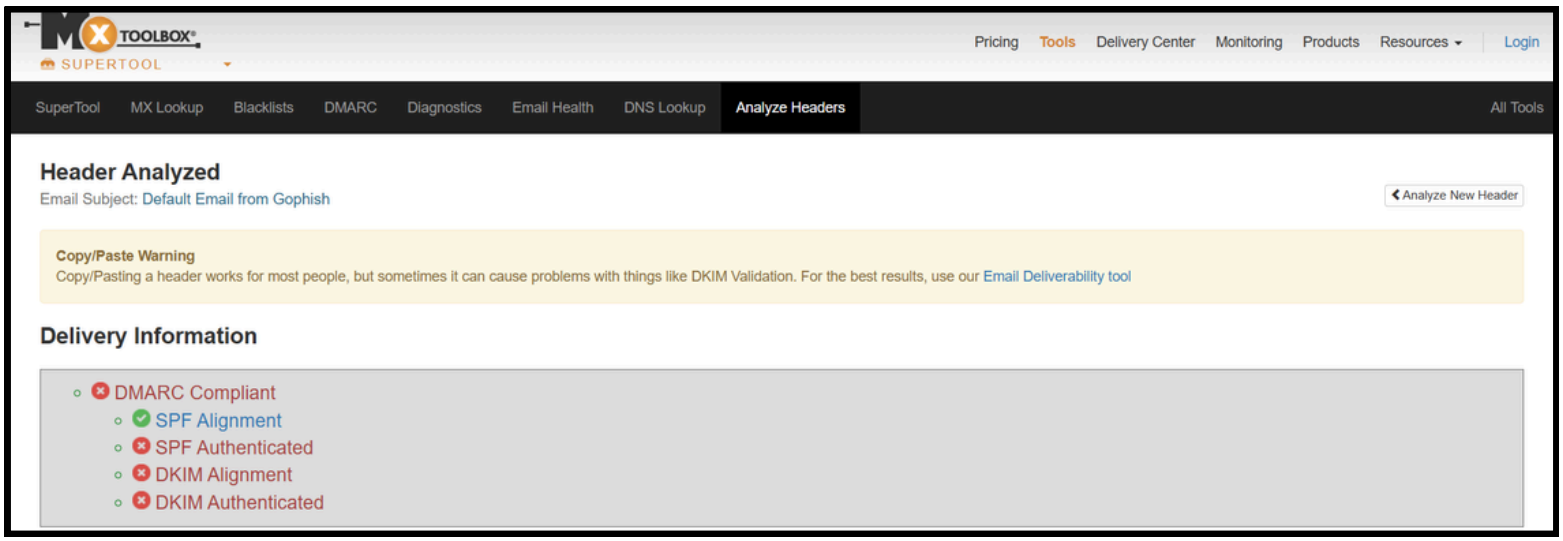

Figure : 03

RECEIVED HEADER

Email originated from **Kali Linux** machine

Sent via : **smtp.gmail.com**

Protocol used: ELTRTPSA (Authenticated SMTP)
Encryption: TLS (AE12_128_GJM_SHA256)

This confirms:

Secure transmission
Authenticated sending via Gmail SMTP
No suspicious relay or unknown server

Received Header

```
Return-Path: **********@gmail.com
Received: from kali ([2409:40f3:2083:a52e:522f:d74b:f919:a ..])
        by smtp.gmail.com with ESMTPSA id d2e1a72fcca58-82f97eb5ce8sm14327884b3a.61.2026.04.22.01.19.23
        for **********@gmail.com
        (version=TLS1_3 cipher=TLS_AES_128_GCM_SHA256 bits=128/128);
        Wed, 22 Apr 2026 01:19:24 -0700 (PDT)
Message-ID: <69e8848c.a70a0220.2de448.bf1b@mx.google.com>
Mime-Version: 1.0
Date: Wed, 22 Apr 2026 04:19:20 -0400
X-Mailer: gophish
Subject: Default Email from Gophish
To: **********@gmail.com
From: fakemail@gmail.com
Content-Type: text/plain; charset=UTF-8
Content-Transfer-Encoding: quoted-printable

It works!

This is an email letting you know that your gophish
configuration was successful.
Here are the details:

Who you sent from: **********@gmail.com

Who you sent to:=20
 First Name: example
 Last Name: example

Now go send some phish!
```

Figure : 04

SPF, DKIM & DMARC ANALYSIS

- DMARC Record

{ v=DMARC1; p=none; sp=quarantine; rua=mailto:mailauth-reports@google.com }

Results:

- DMARC Record Published
- DMARC Policy Not Enforced (p=none)
- Syntax Valid
- External Validation Passed

Important Finding:

- Policy is set to none, meaning no action is taken on failed authentication.
- This reduces protection against spoofed emails.

SPF and DKIM Information

dmarc:gmail.com Hide Solve Email Delivery Problems

v=DMARC1; p=none; sp=quarantine; rua=mailto:mailauth-reports@google.com

Tag	TagValue	Name	Description
v	DMARC1	Version	Identifies the record retrieved as a DMARC record. It must be the first tag in the list.
p	none	Policy	Policy to apply to email that fails the DMARC test. Valid values can be 'none', 'quarantine', or 'reject'.
sp	quarantine	Sub-domain Policy	Requested Mail Receiver policy for all subdomains. Valid values can be 'none', 'quarantine', or 'reject'.
rua	mailto:mailauth-reports@google.com	Receivers	Addresses to which aggregate feedback is to be sent. Comma separated plain-text list of DMARC URIs.

Test	Result	
DMARC Policy Not Enabled	DMARC Quarantine/Reject policy not enabled	More Info
DMARC Record Published	DMARC Record found	
DMARC Syntax Check	The record is valid	
DMARC Multiple Records	Multiple DMARC records corrected to a single record.	
DMARC External Validation	All external domains in your DMARC record are giving permission to send them DMARC reports.	

Reported by **ns2.google.com** on 4/22/2026 at **8:21:36 AM (UTC 0)**, just for you. Transcript

Figure : 05

SECURITY ANALYSIS SUMMARY

COMPONENT	STATUS	REMARKS
Email Authenticity	Valid	Proper Gmail SMTP Authentication
Header Intergrity	Clean	No spoofing detected
Encryption	Secure	TLS encryption used
DMARC Policy	Weak	Not enforced (p = none)
Relay Path	Safe	Suspicious hops

KEY OBSERVATIONS

- Email is legitimate and generated via GoPhish
- No spoofing or malicious activity detected
- Gmail SMTP ensures secure delivery
- Weak DMARC policy could allow phishing in real scenarios

CONCLUSION

The analysis confirms that the email was successfully sent using GoPhish through a secure and authenticated channel provided by Gmail. All headers, relay paths, and authentication mechanisms indicate a legitimate email transmission.

However, the DMARC policy is set to "none", which does not enforce protection against spoofed emails. In a real-world environment, this should be updated to "quarantine" or "reject" to enhance emai security.

Overall, the system is properly configured, but improvements in email authentication policies are recommended to prevent phishing attacks.